Vu Dai An

DevOps Engineer

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EXPERIENCE

Tokyo Tech Lab Viet Nam
Software Engineer — DevOps & Infrastructure Oct. 2024 – Present

- Deliver DevOps and infrastructure work across three products — primary owner of the infrastructure and delivery pipelines for a multi-clinic booking client platform, and supporting the release and environment needs of Kotae (Tokyo Techies' AI chatbot SaaS, on secondment) and StarLive (real-time streaming, multi-vendor client project) in teams of 5–15, working directly with overseas product owners in English.
- Migrated the StarLive admin app to a Turborepo monorepo with shared TypeScript and ESLint config packages, reconfigured SonarQube analysis for the monorepo layout, moved the CI matrix and engine constraint to Node 24, and kept .env files out of Docker build contexts.
- Closed the gap between local, CI, and production environments on the clinic-booking platform — replacing MinIO with LocalStack so tests exercise the same S3 API as the deployed service.
- Documented per-clinic provisioning (Terraform README, new-clinic onboarding guide) and scripted deployment and migration tasks in Bash.

Software Intern Jan. 2024 – Oct. 2024

- Built Gitman, an internal Git repository management tool, with the intern team from the first commit through to company-wide release — Vue.js and TypeScript interface over a Nest.js and MongoDB API.
- Shipped Vue 3 and TypeScript workflow features into the production HRM platform used across the company, then moved onto the Kotae product (on secondment to Tokyo Techies) in the final months of the internship and was promoted to Software Engineer after nine months.

PROJECTS

ERC Booking System Mar. 2026 – Jun. 2026
Role: DevOps Engineer / Backend Developer | Tokyo Tech Lab (client product) | Team of 5
Tech: Terraform · AWS (ECS Fargate, Aurora, ElastiCache, CloudFront, WAF, SQS, SES, Backup) · GitHub Actions · Docker · Liquibase
Clinic scheduling platform architected for per-tenant isolation; primary owner of its infrastructure and CI/CD.

- Primary owner of the Terraform codebase (151 resources, 22 reusable modules): authored the bastion, migration-task, work-hours scaling, GitHub Actions IAM, Route 53, S3 frontend, and SES SMTP modules, plus a per-clinic template that gives each clinic a dedicated production environment.
- Locked the ALB behind CloudFront + AWS WAF (3 managed rule groups, origin-secret header) and served private S3 media via OAC + signed URLs; cut dev compute hours by ~64% with work-hours ECS scaling (60 of 168 hours a week) and no NAT gateway.
- Hardened the API, worker, and migration images (multi-stage, production-only dependencies, non-root, health checks, graceful shutdown) and scoped CI deploy credentials to least privilege (ECR/ECS; S3 + CloudFront).
- Built the GitHub Actions rollout pipeline: patched ECS task definitions, rolling API/worker deploys (100% minimum healthy), old-revision cleanup, path-based selective builds, and an S3 + CloudFront frontend release with immutable hashed assets.
- Engineered a branch-gated Liquibase migration pipeline that runs as a one-off ECS Fargate task per environment and auto-generates CloudWatch Logs links for failure triage.

Kotae Aug. 2024 – Present
Role: Full-Stack Engineer (release support) | Tokyo Techies (sister company, on secondment) | Team of 14
Website: kotae.ai
Tech: AWS Bedrock · OpenAI · Gemini · LangChain · Docker · EKS · Argo CD · Helm · GitHub Actions
AI customer-support chatbot for small businesses: RAG over customer knowledge bases, run on EKS.

- Shipped 21 development deployments across the frontend, backend, and landing services (Nov. 2024 – Jan. 2026) through the team's GitHub Actions to Argo CD + Helm pipeline on EKS.

- Diagnosed crawler failures on the development cluster from worker-pod logs (navigation timeouts across concurrent pods) and raised the crawler's navigation timeout.
- Prototyped a self-hosted Firecrawl stack in Docker Compose (6 services incl. Playwright, Redis, RabbitMQ) with memory limits, health checks, and profiles, and cut crawler concurrency from 10 to 3 pages.
- Extended RAG website ingestion with sitemap.xml URL discovery and feature-flagged OCR of text in crawled images.
- Hardened RAG ingestion against SSRF with private-IP and redirect checks around the crawler.

Portfolio Platform Sep. 2025 – Present

Role: Sole engineer — infrastructure, GitOps, release pipeline, and application
Website: portfolio.anvd.io.vn
Tech: Terraform · AWS (EKS, ECR, ALB, SSM, Lambda, EventBridge) · Cloudflare · Argo CD · Helm · Prometheus · GitHub Actions

Personal site live since Sep. 2025, moved from a VPS to single-node EKS behind Cloudflare in 2026 (not HA; off nights and Sundays).
Code: portfolio-infra · portfolio-gitops · portfolio-web

- Provisioned AWS and Cloudflare in Terraform (67 resource blocks, 6 modules): VPC, EKS with a private API via Session Manager, Pod Identity, ECR, and a Cloudflare-only security group for the controller-managed ALB; CI runs TFLint and Trivy, with actions pinned by SHA.
- Ran the cluster as GitOps: Argo CD ApplicationSets deploy 5 Helm charts (Argo CD, Load Balancer Controller, External Secrets, Prometheus, the app with an HPA), validated with kubeconform.
- Automated releases in GitHub Actions: OIDC to AWS, approval-gated SHA-tagged images in ECR, the tag committed to GitOps, then a version check and smoke test, restoring the last verified tag on failure: 21 deploys in three weeks, ~9 min merge-to-live, one auto-rollback.
- Diagnosed an OOMKilled Argo CD controller (613 Mi sync peak vs. 768 Mi limit) and fixed it with a 1 Gi limit and GOMEMLIMIT; made rollouts drain cleanly with ALB readiness gates and preStop timing.
- Cut Prometheus series by ~88% (~18,900 to 2,356) to fit the Grafana Cloud free tier, allow-listing the 119 metrics behind 18 upstream dashboards; added CloudWatch node-down and scheduler-error alarms.
- Halved node running hours (84 of 168 a week) with a night/Sunday shutdown scheduler (EventBridge + a Python Lambda reconciling every 15 minutes), saving ~18 USD a month (13.5%) at no added cost.
- Wrote 19 ADRs and 7 runbooks, kept in a private workspace repository (available on request).

SKILLS

Infrastructure as Code (IaC) & Scripting: Terraform, Bash, Python
Containers & GitOps: Kubernetes, Helm, GitOps (Argo CD), External Secrets, Docker (BuildKit, Compose)
CI/CD & DevSecOps: GitHub Actions, Jenkins, SonarQube, Trivy, TFLint, Liquibase
Linux & Web Servers: Linux, Nginx, systemd
Cloud (AWS) & Edge: ECS Fargate, EKS, VPC, Aurora, ElastiCache, CloudFront/WAF, SQS, SES, Lambda, EventBridge, IAM, OIDC, Pod Identity, Secrets Manager, Cloudflare
Monitoring & Observability: Prometheus, Grafana (Cloud), CloudWatch (Logs, Alarms)
Databases: PostgreSQL, MySQL, MongoDB, Redis
Development: TypeScript, Node.js / NestJS
AI-assisted Engineering: Claude Code, Cursor

PROFESSIONAL SUMMARY

DevOps Engineer with 2+ years building AWS infrastructure as code with Terraform and delivering containerized workloads on ECS Fargate and EKS. Primary infrastructure owner for a multi-clinic AWS platform with per-clinic environments — 151 Terraform resources across 22 modules, a CloudFront + WAF edge, rolling container rollouts, gated database migrations, and scheduled scaling that cut dev compute hours by ~64%. Runs a self-built EKS platform with GitOps (Argo CD, Helm), Trivy-gated IaC, and automatic release rollback, and supports releases of an LLM-backed AI service on EKS.

EDUCATION

Dai Nam University 2020 – 2024

Bachelor of Science in Information Technology (GPA: 3.58 / 4.0)
Languages: Vietnamese (native), English (professional working proficiency, used daily at work)